GREEN-SYNTHESIZED SILVER NANOPARTICLES AS AN ALTERNATIVE ANTIBACTERIAL STRATEGY AGAINST RESISTANT BACTERIAL PATHOGENS

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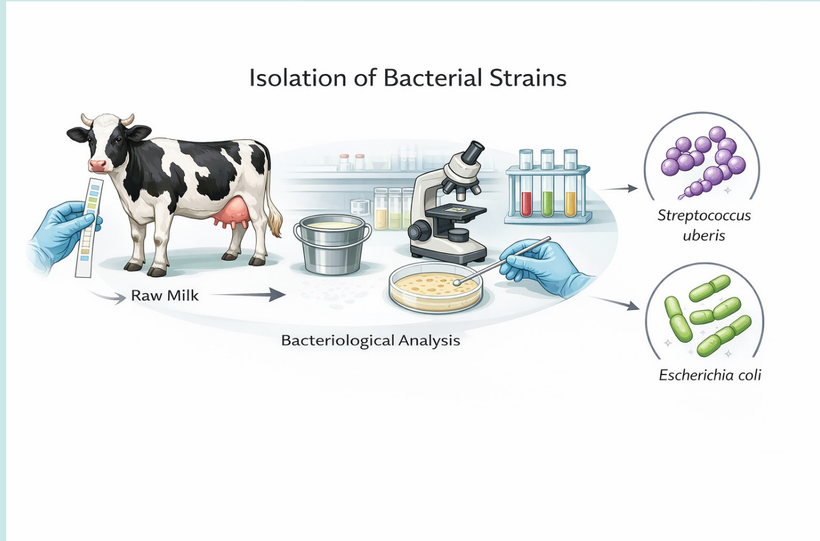
ABSTRACT: Silver nanoparticles synthesized using rosemary (*Rosmarinus officinalis*) extract were evaluated for antibacterial activity against resistant udder pathogens, specifically *Streptococcus uberis* and *Escherichia coli*, isolated from the milk of dairy cows. Using the disk diffusion method, the nanoparticles demonstrated inhibition zones of 10 mm for *E. coli* and 6 mm for *S. uberis*. In contrast, the positive control (gentamicin) showed no effect against these resistant strains, and the negative control (distilled water) exhibited no antibacterial activity. These results suggest that green-synthesized silver nanoparticles hold potential as an alternative or complementary antimicrobial strategy.



MATERIALS AND METHODS

Isolation of bacterial strains

Raw milk samples were collected from mastitic cows with high somatic cell counts (positive California Mastitis Test). Quarter milk samples were analyzed bacteriologically to isolate mastitis-associated pathogens. Resistant ***Streptococcus uberis*** and ***Escherichia coli*** were selected for antibacterial testing.

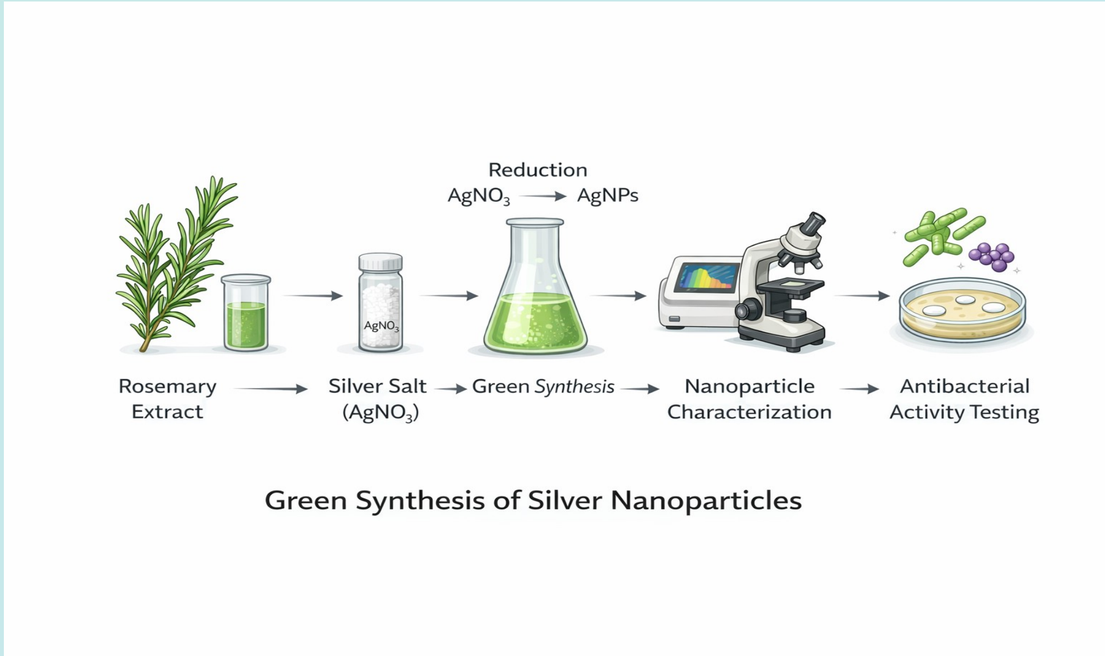
Antibacterial activity assay

Silver nanoparticles synthesized using **rosemary (Rosmarinus officinalis) extract** were tested via the disk diffusion method. Sterile paper disks were loaded with nanoparticle suspension and placed on agar plates inoculated with ***S. uberis*** and ***E. coli***.

Positive control: Gentamicin disks

Negative control: Distilled water disks

Zones of inhibition were measured after incubation.



RESULTS AND DISCUSSION

Rosemary-mediated silver nanoparticles inhibited both bacterial strains from milk. Inhibition zones were **10 mm for *E. coli*** and **6 mm for *S. uberis***. Gentamicin showed no effect, and distilled water was inactive.

The antibacterial activity likely involves multiple mechanisms: disruption of bacterial membranes, generation of reactive oxygen species, and interference with DNA and protein synthesis. Compounds in rosemary extract, such as polyphenols and flavonoids, stabilize the nanoparticles and may enhance their interaction with bacteria.

Differences in susceptibility between *E. coli* and *S. uberis* may reflect structural differences between Gram-negative and Gram-positive bacteria. These findings highlight the potential of green-synthesized silver nanoparticles as an alternative or complementary antimicrobial strategy.

CONCLUSION: Rosemary-mediated silver nanoparticles show promising antibacterial activity against antibiotic-resistant *S. uberis* and *E. coli*. They could serve as an alternative or complementary antimicrobial strategy in veterinary medicine. Further research is needed to optimize dosage, ensure safety, and understand their mechanisms of action.

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